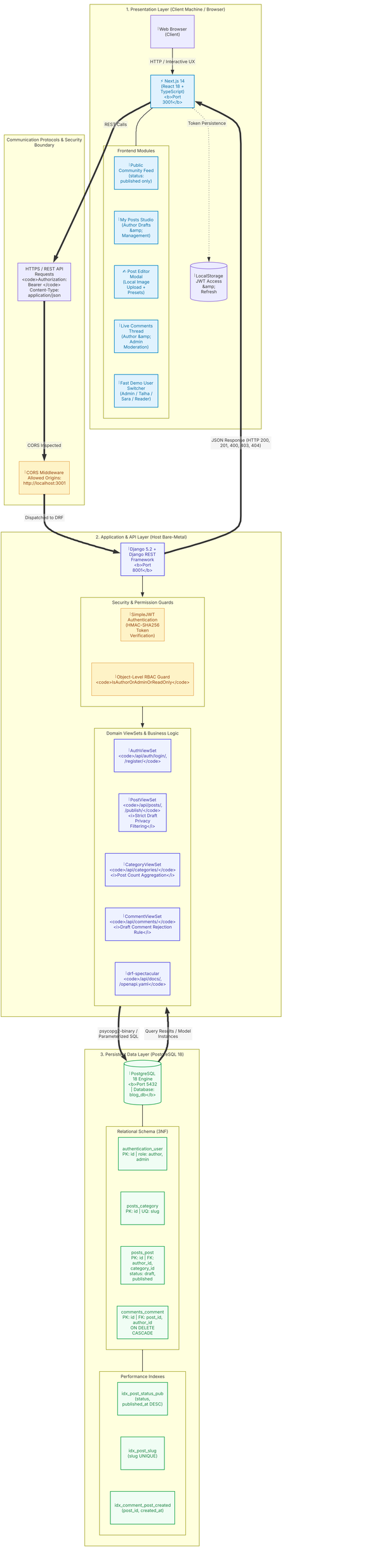

1. Presentation Layer (Client Machine / Browser)
Web Browser (Client)
HTTP / Interactive UX
⚡ Next.js 14 (React 18 + TypeScript) Port 3001
REST Calls
Token Persistence
Communication Protocols & Security Boundary
Frontend Modules
Public Community Feed (status: published only)
My Posts Studio (Author Drafts & Management)
Post Editor Modal (Local Image Upload + Presets)
Live Comments Thread (Author & Admin Moderation)
Fast Demo User Switcher (Admin / Talha / Sara / Reader)
LocalStorage JWT Access & Refresh
HTTPS / REST API Requests <code>Authorization: Bearer </code> Content-Type: application/json
CORS Inspected
CORS Middleware Allowed Origins: http://localhost:3001
Dispatched to DRF
JSON Response (HTTP 200, 201, 400, 403, 404)
2. Application & API Layer (Host Bare-Metal)
Django 5.2 + Django REST Framework Port 8001
Security & Permission Guards
SimpleJWT Authentication (HMAC-SHA256 Token Verification)
Object-Level RBAC Guard <code>IsAuthorOrAdminOrReadOnly</code>
Domain ViewSets & Business Logic
AuthViewSet <code>/api/auth/login/, /register/</code>
PostViewSet <code>/api/posts/, /publish/</code> Strict Draft Privacy Filtering
CategoryViewSet <code>/api/categories/</code> Post Count Aggregation
CommentViewSet <code>/api/comments/</code> Draft Comment Rejection Rule
drf-spectacular <code>/api/docs/, /openapi.yaml</code>
psycopg2-binary / Parameterized SQL
Query Results / Model Instances
3. Persistent Data Layer (PostgreSQL 18)
PostgreSQL 18 Engine Port 5432 | Database: blog_db
Relational Schema (3NF)
authentication_user PK: id | role: author, admin
posts_category PK: id | UQ: slug
posts_post PK: id | FK: author_id, category_id status: draft, published
comments_comment PK: id | FK: post_id, author_id ON DELETE CASCADE
Performance Indexes
idx_post_status_pub (status, published_at DESC)
idx_post_slug (slug UNIQUE)
idx_comment_post_created (post_id, created_at)